

Mini Project

January 10, 2025

1 Library Management System

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[ ]: class Book:
    def __init__(self, title, author):
        self.title = title
        self.author = author
        self.available = True
        self.issued_to = None

    def issue(self, name):
        if self.available:
            self.available = False
            self.issued_to = name
            return True
        return False

    def return_book(self):
        if not self.available:
            self.available = True
            self.issued_to = None
            return True
        return False

    def __str__(self):
        status = "Available" if self.available else f"Issued to {self.
↪issued_to}"
        return f"Title: {self.title}, Author: {self.author}, Status: {status}"

class Library:
    def __init__(self):
        self.books = {}

    def add_book(self, title, author):
        if title in self.books:
            print("This book already exists in the library.")
        else:
            self.books[title] = Book(title, author)
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        print(f"Book '{title}' added successfully.")

def display_books(self):
    print("\nAvailable Books:")
    for book in self.books.values():
        print(book)

def issue_book(self, title, name):
    if title in self.books:
        if self.books[title].issue(name):
            print(f"Book '{title}' has been issued to {name}.")
        else:
            print(f"Sorry, the book '{title}' is currently issued to {self.
↪books[title].issued_to}.")
    else:
        print("This book does not exist in the library.")

def return_book(self, title):
    if title in self.books:
        if self.books[title].return_book():
            print(f"Book '{title}' has been returned successfully.")
        else:
            print(f"The book '{title}' is not currently issued.")
    else:
        print("This book does not exist in the library.")

def view_issued_books(self):
    print("\nIssued Books:")
    for book in self.books.values():
        if not book.available:
            print(f"Title: {book.title}, Issued to: {book.issued_to}")

def login():
    credentials = {"admin": "admin123", "librarian": "lib123"}
    attempts = 0
    while attempts < 3:
        username = input("Enter username: ")
        password = input("Enter password: ")
        if username in credentials and credentials[username] == password:
            print("Login successful!")
            return True
        else:
            print("Invalid credentials. Please try again.")
            attempts += 1
    print("Too many failed attempts. Exiting.")
    return False

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def main_menu(library):
    while True:
        print("\nMenu:")
        print("1. View Books")
        print("2. Add a Book")
        print("3. Issue a Book")
        print("4. Return a Book")
        print("5. View Issued Books")
        print("6. Exit")

        choice = input("Enter your choice: ")

        if choice == '1':
            library.display_books()
        elif choice == '2':
            title = input("Enter the book title: ")
            author = input("Enter the author's name: ")
            library.add_book(title, author)
        elif choice == '3':
            title = input("Enter the book title to issue: ")
            name = input("Enter your name: ")
            library.issue_book(title, name)
        elif choice == '4':
            title = input("Enter the book title to return: ")
            library.return_book(title)
        elif choice == '5':
            library.view_issued_books()
        elif choice == '6':
            print("Exiting the Library Management System. Goodbye!")
            break
        else:
            print("Invalid choice. Please try again.")

if __name__ == "__main__":
    print("Welcome to the Library Management System")
    library = Library()
    if login():
        main_menu(library)

```

Welcome to the Library Management System

Enter username: admin

Enter password: admin123

Login successful!

Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

Enter your choice: 2

Enter the book title: Communism

Enter the author's name: Periyar

Book 'Communism' added successfully.

Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

Enter your choice: 1

Available Books:

Title: Communism, Author: Periyar, Status: Available

Menu:

1. View Books
2. Add a Book
3. Issue a Book
4. Return a Book
5. View Issued Books
6. Exit

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